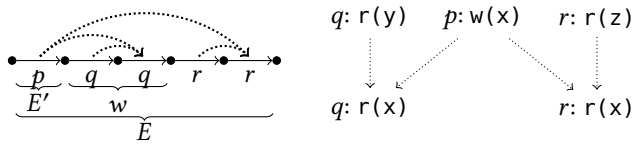

Fig. 4. A sample run of the program in Fig. 1 is shown to the left. This run is annotated by a happens-before relation (the dotted arrows). To the right, the happens-before relation is shown as a partial order. Notice that $E' \vdash q \diamond r$ since q and r are not happens-before related in $E'.r.q$. We also observe that $I_{[E']}(w) = \{q\}$, as q is the only process occurring in w and its first occurrence has no predecessor in the dotted relation in w. Furthermore, $WI_{[E']}(w) = \{q, r\}$, since r is not happens-before related to any event in w.

may lead to a different state: therefore the algorithm must try to explore a corresponding execution sequence.

In Fig. 4, there are two pairs of events e, e' such that $e \lessdot_E e'$, namely $\langle p, 1\rangle, \langle q, 2\rangle$ and $\langle p, 1\rangle, \langle r, 2\rangle$. It also holds for both these pairs that $e \precsim_E e'$ since both q and r are enabled before $\langle p, 1\rangle$. In other words, both the races in the program are reversible.

4 SOURCE SETS

In this section, we define the new concept of source sets. Intuitively, source sets are sets of processes that perform "first steps" in the possible future execution sequences. A set of processes is a source set if any possible future execution sequence contains a "first step" of some process in the source set. We first define two related notions of possible "first steps" in a sequence.

Definition 4.1 (Initials and Weak Initials). For an execution sequence $E.w$, the set $I_{[E]}(w)$ of processes that are *initials* and the set $WI_{[E]}(w)$ of processes that are *weak initials* are defined as follows:

(1) $p \in I_{[E]}(w)$ iff there is a sequence w' such that $E.w \simeq E.p.w'$
(2) $p \in WI_{[E]}(w)$ iff there are sequences w' and v such that $E.w.v \simeq E.p.w'$

The following lemma gives alternative characterizations of the sets $I_{[E]}(w)$ and $WI_{[E]}(w)$.

Lemma 4.2. *For an execution sequence $E.w$ and a process p, we have:*

(1) $p \in I_{[E]}(w)$ iff $p \in w$ and there is no other event $e \in dom_{[E]}(w)$ with $e \rightarrow_{E.w} next_{[E]}(p)$,
(2) $p \in WI_{[E]}(w)$ iff either $p \in I_{[E]}(w)$, or $p \in enabled(s_{[E]})$ and $E \vdash p \diamond w$.

Intuitively, a process in $I_{[E]}(w)$ or $WI_{[E]}(w)$ has no "happens-before" predecessors in $dom_{[E]}(w)$, and is in $I_{[E]}(w)$ if it actually occurs in w.

Proof. We prove each of the cases separately.

(1) (Forward direction) From $E.w \simeq E.p.w'$, using Property (4) of Definition 3.2, we infer $dom_{[E]}(w) = dom_{[E]}(p.w')$, which implies $p \in w$. Furthermore, the fact that p can be moved to the beginning of w implies that p cannot have any predecessors in $dom_{[E]}(w)$, i.e., there is no other event $e \in dom_{[E]}(w)$ with $e \rightarrow_{E.w} next_{[E]}(p)$.
(Reverse direction) If $p \in w$ and there is no other event $e \in dom_{[E]}(w)$ with $e \rightarrow_{E.w} next_{[E]}(p)$, then we can linearize $dom_{[E]}(w)$ in such a way that p occurs first, i.e., there is a w' with $E.w \simeq E.p.w'$.
(2) (Forward direction) From $E.w.v \simeq E.p.w'$, using Property (4) of Definition 3.2, we infer $dom_{[E]}(w.v) = dom_{[E]}(p.w')$. If $p \in w$, then the events in v can be removed from w' to obtain $dom_{[E]}(w) = dom_{[E]}(p.w'')$ for some w'', implying $p \in I_{[E]}(w)$. If $p \notin w$, then $p \in v$.